# Lecture 2 v2:
# Review of Statistics

25117 - Econometrics

Universitat Pompeu Fabra

September 30th, 2024

# What we learned in the last lesson

- Econometrics quantifies economic relationships
- Randomized control experiments are often too expensive or unethical. We have to rely on observational data that is potentially jointly determined.
- The probabilities with which a random variable takes on different values are summarized by the cumulative and probability distribution functions
- The expected value of a random variable $Y$, $\mathbb{E}(Y)$, is its probability-weighted average value. The variance is $\mathbb{E}[(Y - \mathbb{E}(Y))^2]$.
- The joint probabilities for two random variables, $X$ and $Y$, are summarized by their joint probability distribution. The conditional probability distribution of $Y$ given $X = x$ is the probability distribution of Y, conditional on X taking on the value $x$.

# What we learned in the last lesson

- Simple random sampling produces $n$ random observations that are i.i.d.
- The sample average, $\overline{Y}$ is a random variable with a sampling distribution. If observations are i.i.d. then,
  1. $\overline{Y}$ has mean $\mu_Y$ and variance $\sigma^2_{\overline{Y}} = \frac{\sigma^2_Y}{n}$;
  2. the LLN says that $\overline{Y}$ converges in probability to $\mu_Y$
  3. the CLT says that $(\overline{Y} - \mu_Y)/\sigma^2_{\overline{Y}} \sim \mathcal{N}(0;1)$ when $n$ is large
- The sample average, $\overline{Y}$ is a BLUE estimator of the population mean. If observations are i.i.d. then,
  1. $\overline{Y}$ is unbiased;
  2. by the LLN, $\overline{Y}$ is consistent
  3. by the CLT, $\overline{Y}$ has an approximate normal sampling distribution when $n$ is large
  4. $\overline{Y}$ is efficient – it has the smallest sampling variance of all linear unbiased estimator

# What is Hypothesis Testing?

**Hypothesis Testing** is a statistical method used to make decisions about population parameters based on sample data.

- We start with two competing hypotheses:
  - Null Hypothesis ($H_0$): The default assumption, no effect, no difference, etc.
  - Alternative Hypothesis ($H_1$): The statement we want to test, suggests an effect or difference.
- We collect sample data and use it to make an informed decision about which hypothesis is more likely.
- For example, if we want to test that a population mean takes a specific value, we write $H_0 : \mu_Y = \mu_Y^0$
- Here the alternative hypothesis would be $H_1 : \mu_Y \neq \mu_Y^0$. We talk about two-sided alternatives because $\mu_Y$ may be greater or smaller than $\mu_Y^0$

# Why Use Hypothesis Testing?

Hypothesis testing helps us:

- Make data-driven decisions.
- Evaluate claims or theories.
- Determine if observed differences are statistically significant or just due to chance.
- Assess the reliability of our results.

It provides a structured approach for drawing conclusions from data.

**Example:** A pharmaceutical company develops a new drug to lower blood pressure.

- Hypotheses:
  - Null Hypothesis ($H_0$): The new drug has no effect on blood pressure.
  - Alternative Hypothesis ($H_1$): The new drug is effective in lowering blood pressure.
- Clinical trial data is collected to assess the drug's efficacy.
- Hypothesis testing determines if the observed blood pressure reduction is statistically significant.
- Outcome: Approval for medical use if the null hypothesis is rejected.

# Key Concepts

- **Null Hypothesis ($H_0$)**: The default hypothesis, often representing no effect or no difference.
- **Alternative Hypothesis ($H_1$)**: The statement we want to test, suggesting an effect or difference.
- **Significance Level ($\alpha$)**: The threshold for determining statistical significance. Common choices are 0.05 or 0.01.
- **Test Statistic**: A numerical summary of the data used in hypothesis testing.
- **P-value**: The probability of obtaining results as extreme as the observed data, assuming the null hypothesis is true.

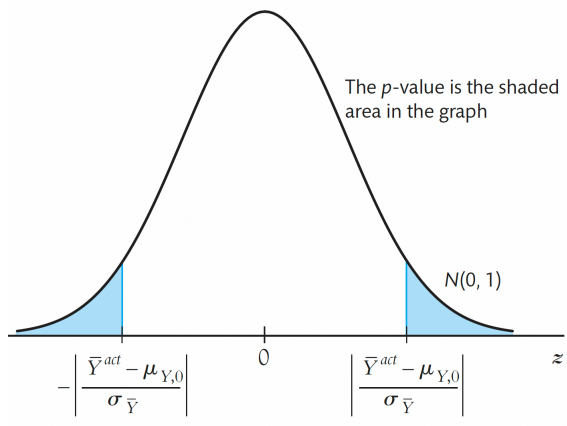
These concepts are fundamental in hypothesis testing and guide our decision-making process.

# Computing the P-Value

- Mathematically, one can write the p-value

$$Pr_{H_0}[\mid \overline{Y} - \mu_Y^0 \mid > \mid \overline{Y}^{act} - \mu_Y^0 \mid]$$

- Computing the p-value, therefore, requires to know the sampling distribution of $\overline{Y}$
- When $n \to \infty$, we know that $\frac{\overline{Y} - \mu_Y}{\sigma_{\overline{Y}}} \sim \mathcal{N}(0,1)$, i.e., under the null hypothesis, $\frac{\overline{Y} - \mu_Y^0}{\sigma_{\overline{Y}}} \sim \mathcal{N}(0,1)$
- But we do not always know $\sigma_Y$ to compute $\sigma_{\overline{Y}}$...



# Sample Variance and Sample Standard Deviation

- The sample variance, $s_Y^2$ is an estimator of the population variance $\sigma_Y^2$

$$s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2$$

- The sample standard deviation, $s_Y = \sqrt{s_Y^2}$, is an estimator of the population standard deviation $\sigma_Y$
- There are two differences with the population variance, $\mathbb{E}[(Y-\mu_Y)^2]$:
  1. $\mu_Y$ is unknown, so it is replaced by its natural estimator $\overline{Y}$
  2. We divide by $n-1$ (instead of $n$) to correct for the small downward bias generated by estimating $\mu_Y$ with $\overline{Y}$ (Bessel's correction)
- We can show that $s_Y^2 \xrightarrow{p} \sigma_Y^2$ as $n \to \infty$

# Standard Error to compute the p-value

- Because the variance of the sampling distribution, $\sigma^2_{\overline{Y}} = \frac{\sigma^2_Y}{n}$, and sample variance is a consistent estimator of the population variance, $s^2_Y \xrightarrow{p} \sigma^2_Y$, then $\frac{s^2_Y}{n} \xrightarrow{p} \frac{\sigma^2_Y}{n} = \sigma^2_{\overline{Y}}$
- In other words, the standard error, $SE(\overline{Y}) = \hat{\sigma}_{\overline{Y}} = \frac{s_Y}{\sqrt{n}}$ is an estimator of the standard deviation of the sampling distribution of $\overline{Y}$
- Then, when $\sigma_{\overline{Y}}$ is unknown because $\sigma_Y$ is unknown, one can still compute the p-value (under the assumption that $Y_1 \ldots Y_n$ are i.i.d.):

$$\text{p-value} = 2\Phi\left(-\left|\frac{\overline{Y}^{act} - \mu_Y^0}{SE(\overline{Y})}\right|\right)$$

where the Standard Normal Cumulative Distribution Function, denoted as $\Phi(z)$, is a mathematical function that gives the probability that a standard normal random variable is less than or equal to a specific value $z$.

$$\Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

# The t-statistic and its use for hypothesis testing

- The standardized sample average is called t-statistics, t-ratio, or Student's t-statistic.

$$\text{t} = \frac{\overline{Y} - \mu_Y^0}{SE(\overline{Y})}$$

- Because $s_Y^2 \xrightarrow{p} \sigma_Y^2$ as $n \to \infty$, then $t \sim \mathcal{N}(0,1)$ as $n \to \infty$
- In general, we say we reject $H_0$ (or that $\overline{Y}$ is significantly different from $\mu_Y^0$) if the p-value $< .05$ (or when $\alpha = .01$)
- To summarize, testing $H^0$ against $H^1$ involves:
  1. Compute the standard error of $\overline{Y}$, $SE(\overline{Y})$
  2. Compute the corresponding t-statistic
  3. Compute the p-value. Reject $H^0$ at the 5% significance level if the p-value is less than 0.05 (or alternatively, if $\mid t^{act} \mid > 1.96$)
- But what are the consequences of making errors in hypothesis testing?

# Type I and Type II Errors in Hypothesis Testing

## Type I Error (False Positive)

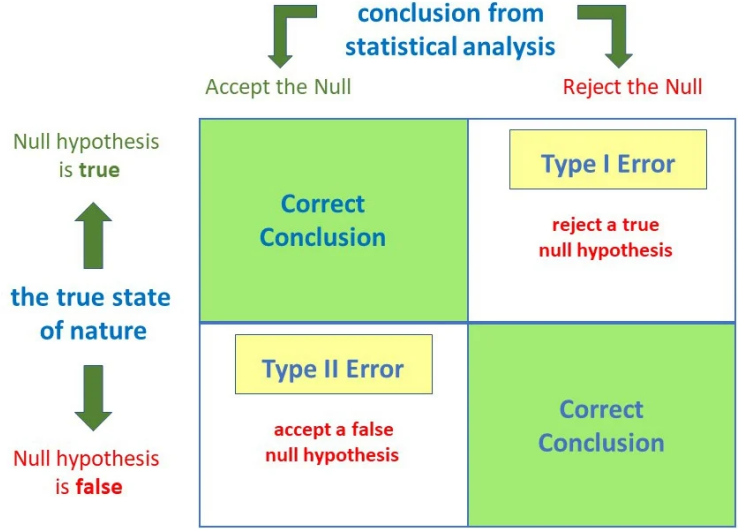
- Occurs when we reject a null hypothesis that is actually true.
- Denoted as $\alpha$ (alpha), the chosen significance level.
- Probability of Type I Error: $\alpha$.
- Example: Incorrectly concluding a drug is effective when it's not.
- Consequence: Often considered more serious due to potential harm.

## Type II Error (False Negative)

- Occurs when we fail to reject a null hypothesis that is actually false.
- Denoted as $\beta$ (beta).
- Probability of Type II Error: Influenced by factors like sample size, effect size, and $\alpha$.
- Example: Failing to detect a real positive effect of a drug.
- Consequence: May result in missed opportunities for important discoveries or interventions.

In hypothesis testing, the choice of $\alpha$ and sample size affects the balance between Type I and Type II errors.

# Errors in Hypothesis Testing



# $H^0$: Patient is *not* pregnant | $H^1$: Patient is pregnant



# What are Confidence Intervals (CIs)?

## Definition

A Confidence Interval (CI) is a range of values constructed from sample data that is likely to contain the true population parameter with a certain level of confidence.

## Example

If we estimate the average height of a population to be 170 cm with a 95% CI to be (165 cm, 175 cm), it means we are 95% confident that the true average height falls within this interval.

- They provide a range of plausible values for a population parameter, not just a single point estimate.
- CIs help us quantify the uncertainty in our estimates.
- CIs help us make informed decisions in the presence of uncertainty.
- They are widely used in research, business, and policy to convey the reliability of estimates.

# Computing CIs

- 90% confidence interval for $\mu_Y = \overline{Y} \pm 1.64 \times SE(\overline{Y})$
- 95% confidence interval for $\mu_Y = \overline{Y} \pm 1.96 \times SE(\overline{Y})$
- 99% confidence interval for $\mu_Y = \overline{Y} \pm 2.58 \times SE(\overline{Y})$

## Example

- You collect a random sample of $n = 50$ individuals at UPF and measure their heights. The sample mean height, $\overline{Y}$, is 170 cm, and the sample standard deviation, $s_Y$, is 8 cm.
- The standard error, $SE(\overline{Y})$, is $\frac{8}{\sqrt{50}}$
- The 95% confidence interval is given by

$$170 \pm 1.96 \times \frac{8}{\sqrt{50}} \approx [167.78; 172.22]$$

- We are 95% confident that the true average height of adults at UPF falls within the interval [167.78 ; 172.22]

# Mean Difference Testing: Comparing Groups 1/2

- Consider the null hypothesis that mean earnings for these two populations differ by a certain amount, say, $d_0$

$$H^0 : \mu_Y^a - \mu_Y^b = d_0 \quad \textbf{vs} \quad H^1 : \mu_Y^a - \mu_Y^b \neq d_0$$

- In a lot of applied studies $d_0 = 0$, which means

$$H^0 : \mu_Y^a = \mu_Y^b \quad \textbf{vs} \quad H^1 : \mu_Y^a \neq \mu_Y^b$$

- You have computed $\overline{Y}^a$ and $\overline{Y}^b$. From there, derive the sample variance $var(\overline{Y}^a - \overline{Y}^b)$ and deduce the $SE(\mu_Y^a - \mu_Y^b)$

$$\begin{aligned} var((\overline{Y}^a - \overline{Y}^b) - d_0) &= var(\overline{Y}^a) + var(\overline{Y}^b) - 2cov(\overline{Y}^a, \overline{Y}^b) \\ \Rightarrow \quad SE(\overline{Y}^a - \overline{Y}^b - d_0) &= \sqrt{\frac{var(\overline{Y}^a)}{n} + \frac{var(\overline{Y}^b)}{n}} \end{aligned}$$

# Mean Difference Testing: Comparing Groups 2/2

- Once we have $SE((\mu_Y^a - \mu_Y^b) - d_0)$, we can easily compute the corresponding t-statistic

$$t = \frac{(\overline{Y}^a - \overline{Y}^b) - d_0}{SE((\overline{Y}^a - \overline{Y}^b) - d_0)}$$

$\rightarrow$ We can reject that $\mu_Y^a - \mu_Y^b = d_0$ at the 5%-level when $\mid t \mid > 1.96$

- Similarly, we can construct the confidence interval for $\mu_Y^a - \mu_Y^b = d_0$

$$CI_{\mu_Y^a - \mu_Y^b = d_0} = (\overline{Y}^a - \overline{Y}^b) + 1.96 \times SE((\overline{Y}^a - \overline{Y}^b) - d_0)$$

$\rightarrow$ We can reject that $\mu_Y^a - \mu_Y^b = d_0$ at the 5%-level when $d_0 \notin CI_{\mu_Y^a - \mu_Y^b = d_0}$

- Comparing mean differences is very useful, e.g., when one wants to verify that control and treatment groups are comparable in means before receiving treatment (i.e., balance tests).

# Causal Inference in Randomized Control Trials

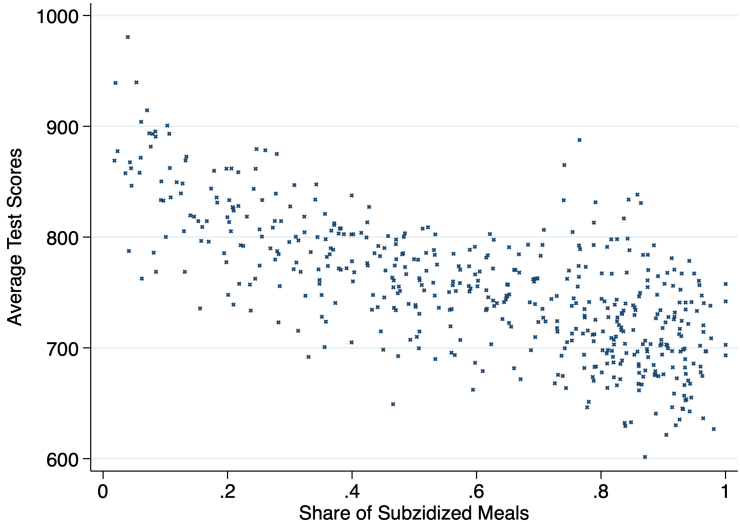
- In the ideal, golden standard RCT case, the causal effect of a treatment variable $D$ on an outcome variable $Y$ is the difference in conditional expectations; i.e., for a dummy treatment,

$$\mathbb{E}(Y \mid D = 1) - \mathbb{E}(Y \mid D = 0)$$

- The randomized assignment of the treatment helps ensure that any observed differences are likely due to the intervention, reducing confounding.
- Mean Difference Testing allows us to test whether $\mathbb{E}(Y \mid D = 1) = \mathbb{E}(Y \mid D = 0)$
- Example 1: Drug Efficacy
  - Treatment Group: Receives the new drug.
  - Control Group: Receives a placebo.
  - Mean Difference Testing helps determine if the drug has a significant effect on health outcomes.
- Example 2: Educational Interventions
  - Different teaching methods in classrooms.
  - Mean Difference Testing assesses if one method leads to better student performance.

# The California Test Score Data Set

- All K-6 and K-8 California school districts ($n = 500$) (CPS, 1999)
- Test score: sum of math and English/language arts, 5th grade
- Subsidized Meals: free or reduced-price meals



# The California Test Score Data Set

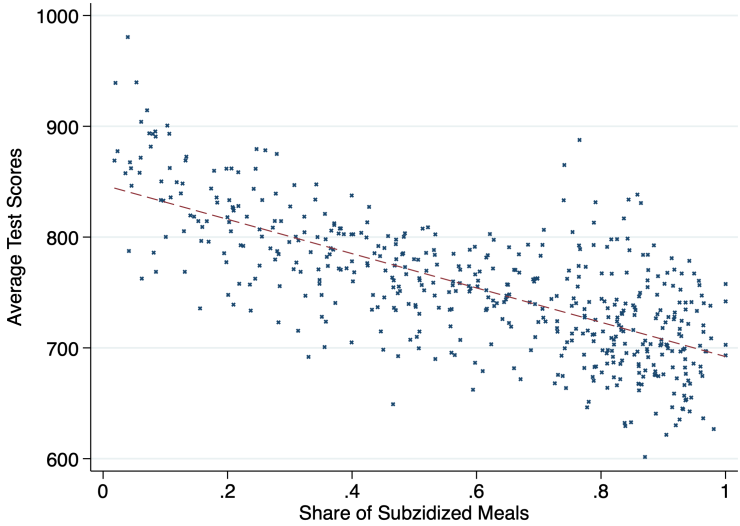
The scatter plot does not tell us much about each variable, independently. We can provide descriptive statistics of the sample:

| | Median | p25 | p75 | Min | Max |
|---|---|---|---|---|---|
| *Test Scores* | 751.5 | 708.95 | 791.75 | 601.7 | 980.7 |
| *Share of Subsidized Meals* | .6511 | .3755 | .8432 | .018 | 1 |

| | Mean | sd | Skewness | Kurtosis |
|---|---|---|---|---|
| *Test Scores* | 753.8542 | 60.3035 | .3696458 | 3.150585 |
| *Share of Subsidized Meals* | .6018114 | .2758199 | -.445432 | 1.958609 |

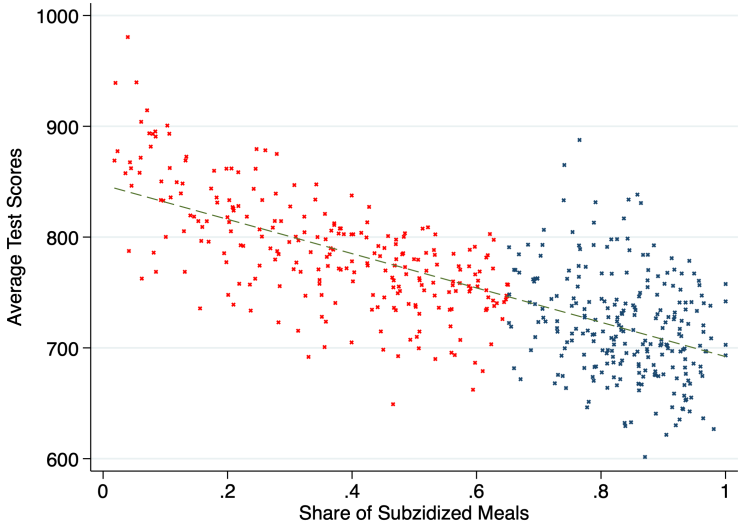
# The California Test Score Data Set

- This allows us to study the joint distribution of the *Test Scores* and the *Share of Subsidized Meals*
- We can compute the sample covariance, $s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})$
- Here, $s_{XY} = -11.7839$
- The sample correlation, $r_{XY} = \frac{s_{XY}}{s_X s_Y}$
- Here, $r_{XY} \approx \frac{-11.78}{60.3 \times 0.28} \approx -.7$



# The California Test Score Data Set

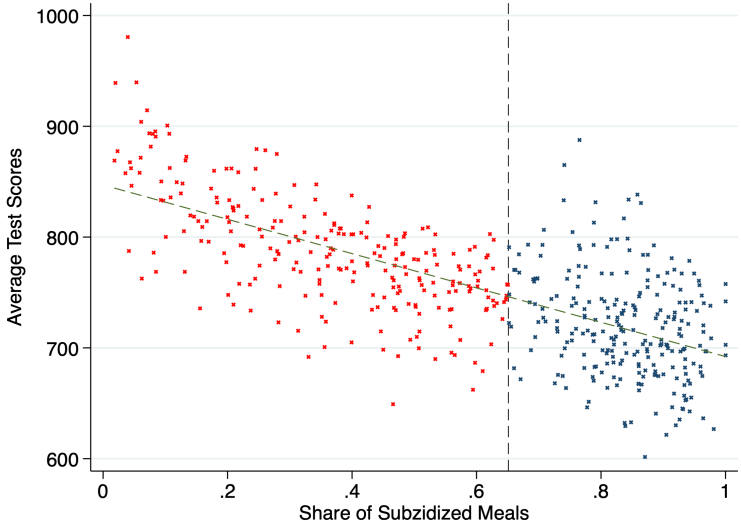
- Are schools with less subsidized meals significantly better than schools with more subsidized meals?
- We can split the distribution into two groups:
  1. Below-median *Share of Subsidized Meals* (red)
  2. Above-median *Share of Subsidized Meals* (red)



# The California Test Score Data Set

Estimation:

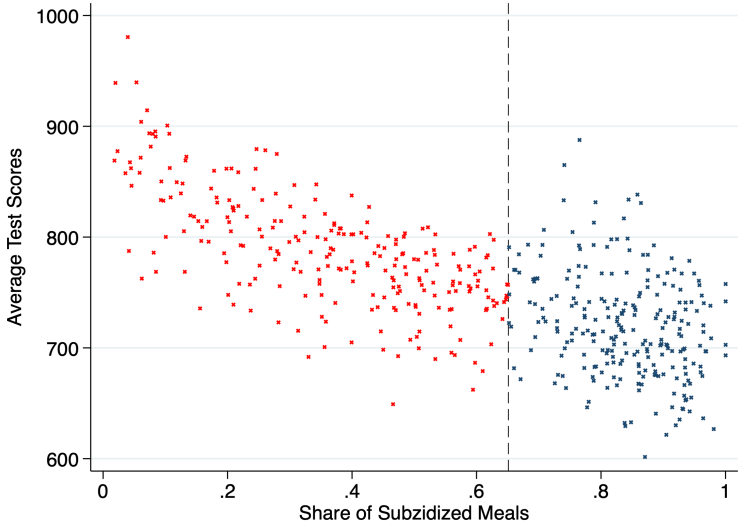
- Compare average test scores in districts with low *Share of Subsidized Meals* to those with high *Share of Subsidized Meals*



# The California Test Score Data Set

Hypothesis testing:

- Test the "null" hypothesis that the mean test scores in the two types of groups are the same, against the "alternative" hypothesis that they differ
- $H^0 : \overline{Y}^{X^L} - \overline{Y}^{X^H} = 0$
- $H^1 : \overline{Y}^{X^L} - \overline{Y}^{X^H} \neq 0$

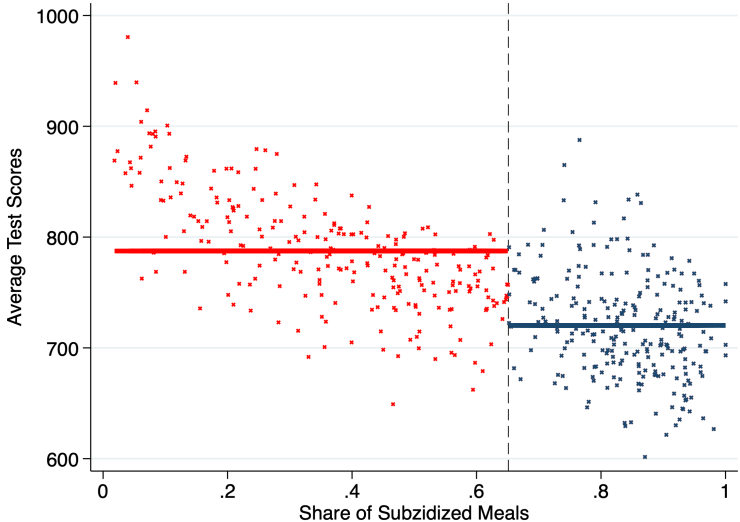


# The California Test Score Data Set

We compute the descriptives for each group

| | $\overline{Y}$ | $s^2$ | $n$ |
|---|---|---|---|
| $X^L$ | 787.46 | 52.62 | 250 |
| $X^H$ | 720.24 | 47.44 | 250 |

- $\overline{Y}^{X^L} - \overline{Y}^{X^H} = 67.214$
- $SE(\overline{Y}^{X^H} - \overline{Y}^{X^L}) = \sqrt{\frac{47.44^2}{250} + \frac{52.62^2}{250}} \approx 4.48$
- $t = 67.214/4.48 \approx 15 > 1.96$

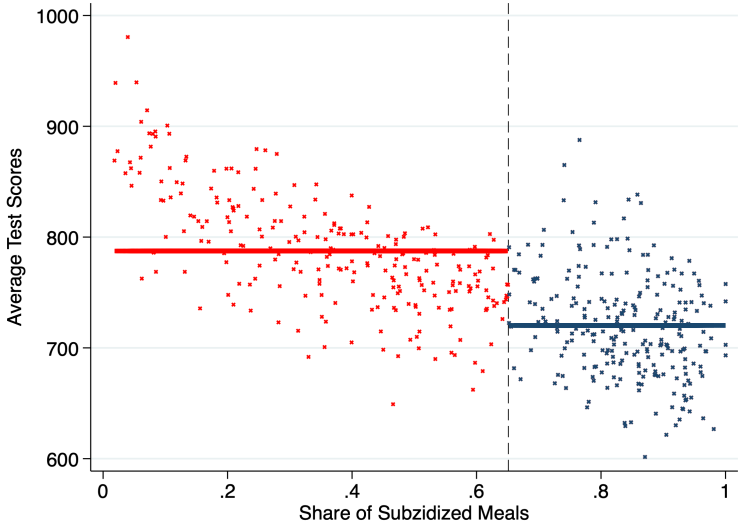


# The California Test Score Data Set

Alternatively, we can also compute the 95% CI for the difference

$$\begin{aligned}\overline{Y}^{X^L} - \overline{Y}^{X^H} &\pm 1.96 \times SE(\overline{Y}^{X^H} - \overline{Y}^{X^L}) \\ &= 67.214 \pm 1.96 \times 4.48 \\ &= [58.41; 76.018] \\ &\notin 0\end{aligned}$$

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# Material I

– **Textbooks:**
  - Introduction to Econometrics, 4th Edition, Global Edition, by Stock and Watson – Chapter 3